



Development of Sustainable Okra Fiber-Nano-SiO₂-reinforced Epoxy Composites: Mechanical Characterization and Machine Learning-based Prediction

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ABSTRACT

The transformation to environmentally friendly components in composite production is essential to minimize the environmental impact and expenditures. Substituting natural fibers, especially those from plants like okra, for conventional reinforcing elements in polymer composites makes them eco-friendlier and more cost-effective. This study aims to characterize and produce polymer composites reinforced with nano-SiO₂ filler and chemically-treated okra fibers. Okra fibers, derived from the okra plant, stand out due to their accessibility, mechanical qualities, and positive impact on the environment. Physical and mechanical properties of composites, such as water absorption, tensile strength, and flexural strength, are studied concerning weight percentage, length, thickness, and nano-SiO₂ content. Composites with fibers of 12 mm length, 2 mm thickness, and 8 wt.% nano-SiO₂ arrangement have shown excellent performance. These composites have a flexural strength of 55.65 MPa and an ultimate tensile strength of 35.92 MPa, which enhances the structural integrity and longevity of the composites. Increasing the fiber weight percentage results in higher water absorption rates, reaching a peak of 3.26%, which highlights the significant impact of fiber dimensions. In addition, when comparing the performance of Response Surface Methodology (RSM) and Feed-Forward Artificial Neural Network (FFANN), predictive modeling using RSM yielded an average accuracy of 85–90%, whereas FFANN achieved 95–98%. Finally, the experimental results were validated through microstructural studies, which underscored the potential of okra fibers to enhance the performance of natural fiber-reinforced composites in various industrial applications.

Keyword: Okra fibers; Flexural strength; Mechanical properties; Water absorption; Artificial neural network.

1. INTRODUCTION

The rising demand for sustainable materials in composite manufacturing has resulted in improved interest in natural fiber-reinforced polymer composites as environmentally benign substitutes for traditional synthetic systems. Bast fibers, including jute, hemp, and flax, exhibit remarkable promise among natural fibers. Nevertheless, underutilized fibers such as okra (*Abelmoschus esculentus*), an easily accessible agricultural by-product, have high tensile strength, low density, and favorable mechanical properties upon surface modification, thereby establishing them as potential candidates for bio-composite reinforcement. Recent improvements highlight the importance of fiber reinforcement and the integration of fillers to augment interfacial bonding, reduce void content, and boost thermal and mechanical performance. In this context, nano-SiO₂ is regarded as one of the most effective nano-reinforcements for natural fiber composites due to its ability to mitigate poor matrix adhesion and elevated moisture susceptibility. Kumar *et al.* (2021a) explored the mechanical and physical properties of composites

made from natural leaf fibers and discussed the effects of fiber loading, length, and silane treatment on performance. Natural leaf fibers offer eco-friendly advantages, better insulation, and easier processing. Their reinforcement improves composite sustainability and mechanical strength.

Sunhemp and flax fibers were blended with polyester resin to fabricate hybrid composites with varied fiber lengths and orientations. Triaxial-oriented fibers exhibited the best tensile (86.3 MPa), flexural (117.1 MPa), and impact (153.8 kJ/m²) strengths. Orientation and fiber length had major effects on mechanical performance (Krishna *et al.* 2022). Agro-waste fibers from banana, sisal, and date palm were used to fabricate composites with varied fiber lengths (6–10 mm). Testing showed mechanical properties improved with longer fibers and hybridization. Water absorption tests confirmed suitability for hydrophobic environments (Kumar *et al.* 2021b). Caryota fibers from fishtail palms were used to make composites showing strong mechanical properties. The composite achieved a maximum tensile strength of 72.24 MPa (Ramkumar *et al.* 2024).

Optimization of flax fiber composites with nano-TiO₂ fillers and cryogenic treatment was achieved using RSM. Tensile, flexural, and interlaminar shear strengths (ILSS) improved with 300 gsm fiber, 5% nano-TiO₂, and 15-min cryogenic exposure (Mohammed *et al.* 2024). Glass fiber-reinforced epoxy composites with different fiber forms (bi-directional and random) were tested for tensile and bending properties. Random fibers showed maximum tensile strength, while thin bi-directional mats offered the highest ductility (Alhassany *et al.* 2020). Hemp fiber nanocomposites reinforced with graphene powder were developed and analyzed for flexural strength and ILSS. Best properties were achieved with 300 gsm hemp and 6 wt.% graphene. ANN modeling predicted mechanical behavior with high accuracy ($R^2 \approx 0.97$) (Natrayan *et al.* 2023). Increased coconut fiber loading improved tensile, flexural, and impact strength initially, but excessive loading reduced performance. Alkali treatment enhanced all mechanical properties. SEM analysis confirmed improved fiber-matrix bonding (Kaliappan and Natrayan, 2023). Pineapple leaf (PALF) and sunhemp (SH) fiber-reinforced epoxy composites were fabricated and studied. NaOH treatment improved tensile, compression, flexural strength, density, and water absorption, although impact strength was slightly reduced. SEM showed better fiber-matrix adhesion after treatment (Premnath, 2022). *Prosopis juliflora* bark, banana fiber, and coconut fiber were used to fabricate epoxy hybrid composites. Untreated samples showed strength improvements up to 25% coconut loading, beyond which properties reduced. Alkali treatment further enhanced tensile, flexural, impact, and hardness values (Kumar and Raja, 2021c).

Natural bamboo fiber reinforced epoxy composites were fabricated with varying nano-SiO₂ filler loadings (0, 2, 4 wt.%). Mechanical tests (tensile, flexural, impact, hardness) and water absorption were assessed. Best results were observed at moderate nano-SiO₂ levels, while excessive filler reduced strength (Kumar *et al.* 2023). Peepal fiber-based epoxy composites were fabricated with SiO₂ and SiC nanofillers for reinforcement. The epoxy/peepal/2 wt.% SiO₂ composite showed the highest tensile (88 MPa), flexural (72 MPa), and impact strengths (9.5 kJ/m²) and hardness (Vignesh *et al.* 2024). Mechanical properties of bamboo fiber composites were enhanced by adding silicon carbide (SiC) fillers through hand lay-up fabrication. Water absorption resistance also improved. SEM analysis confirmed better fiber-matrix bonding with filler addition (Tahir *et al.* 2023). Plantain fiber and MWCNT-reinforced epoxy nanocomposites were optimized for impact strength using RSM and ANN methods. KMnO₄-acetone surface treatment improved fiber-matrix adhesion (Imoisili *et al.* 2023). This study modeled the compressive and tensile strengths of jute fiber-reinforced concrete composites using RSM, ANN, and SVR (Support Vector Regression) methods. Data indicated that SVR provided the most accurate predictions.

Properties depended on water-cement ratio, fiber length, and fiber volume (Sultana *et al.* 2020). Abrasive Waterjet Machining (AWJM) optimization was performed on Roselle fiber composites filled with eggshell powder. An ensemble ML model combined k-NN, RF, LSTM, and ANN with RSM to optimize machining parameters. Best results achieved minimal surface roughness (3.34 μm) and kerf angle, improving machining precision for sustainable composites (Periyappillai *et al.* 2025).

The bending mechanical properties of polymer composites reinforced with Diss, Sisal, and Luffa natural fibers at different filling densities (10–25%) were analyzed. Using fiber analysis, FTIR, and Artificial Neural Network (ANN) modeling, they found that 25% fiber content provides the highest stress values, particularly for Diss fiber at 25.61 MPa. ANN models showed superior prediction capabilities compared to RSM, with correlation coefficients exceeding 0.99 (Saada *et al.* 2024). Washingtonia mesh waste (WMW) bio-mortars reinforced with alkali-treated natural fibers were investigated. RSM and ANN were applied to optimize NaOH concentration, immersion time, and WMW content; the results showed that 1% cement replacement with 2.5% NaOH-treated WMW after 4 hours provided improved mechanical properties (Khelifi *et al.* 2024).

The mechanical, thermal stability, and water absorption behavior of silane-treated high-content cellulose okra fiber (CCO) reinforced with tamarind kernel powder (MTP) toughened epoxy composites were examined. SEM images confirmed enhanced particle dispersion, and the treated composites exhibited minimal water absorption (0.13%), indicating superior hydrophobicity suitable for construction, military, and domestic applications (Sivakumar *et al.* 2023). The authors investigated silane-treated high-cellulose okra fiber (CCO) and tamarind kernel powder (MTP)-toughened epoxy composites in terms of their mechanical, thermal, and water resistance properties. (Nandakumar *et al.* 2024). Authors focused on improving the surface finish in okra fiber epoxy composites by incorporating 30% *Acacia catechu* filler. The filler-reinforced composites exhibited a reduced surface roughness of 4.07 μm compared to pure okra composites, with a significant p-value of 0.024 (Saikrishna and Bharathiraja, 2023). For composites made by reinforcing epoxy resin with date seed powder and okra fiber, the mechanical tests revealed that combining 15% filler and 20% fiber yielded optimum properties, notably enhancing tensile, flexural, and impact strengths (Pillai *et al.* 2023). A natural fiber-based fiber metal laminate (FML) using okra fiber mats, aluminum 2024-T3 foils, and rice husk-derived biosilica. With surface treatments, such as sandblasting and silanization, composites exhibited a storage modulus of 4.2 GPa and energy release rates up to 0.76 MJ/m² (Ramesh *et al.* 2022).

The discontinuous casting method is employed to fabricate polymer composites reinforced with okra fiber in this investigation. By controlling fiber length, fiber content, thickness, and nano-SiO₂ concentration during production, this technique was selected to enhance the precision of mechanical and physical property optimization. Water absorption rate, flexural strength, and tensile strength are the variables examined in this study. In comparison to conventional methods that use continuous fibers, the hand lamination approach allows for the consistent dispersal and saturation of intermittent fibers, resulting in a more efficient interaction between the two. The novel aspect of this study is the utilization of okra fibers as environmentally friendly reinforcements for epoxy composites using the intermittent casting technique. This technology provides a cost-effective, adaptable, and efficient method for incorporating natural fibers into polymer matrices. This study shows that optimizing the percentage of fiber weight, length, thickness, and nano-SiO₂ content greatly enhances the mechanical properties of the composite. Automotive, construction, and packaging are just a few examples of industries that could benefit from these lightweight, high-strength materials. These composites are highly attractive when considering eco-friendly manufacturing options, due to the environmental friendliness of the okra fibers made from agricultural waste.

2. EXPERIMENTAL METHODOLOGY

2.1 Materials

This experimental work utilized epoxy (LY556; density: 1.14 g/cm³, viscosity: 926 mPa·s) as the matrix material and hardener (HY951; density: 0.90 g/cm³, viscosity: 60 mPa·s) in 10:1 ratio (Mayakannan *et al.* 2023). Nano-silicon dioxide (nSiO₂) filler, with a density of 2.2 g/cm³ and a particle size of 20 nm, is used. The epoxy matrix was provided by Kovai Cheenu Enterprises, Coimbatore, India. Okra fiber reinforcement was obtained from the Erode district in Tamil Nadu, India, and the nanomaterials were provided by Matrics Enterprises, Nagercoil, India.

2.2 Methodology

Fig. 1 illustrates the method of extracting okra fibers. The biomass of a complete okra plant was initially taken and subsequently categorized into its roots, stem, and leaves. After the separation of the stems, they were subjected to a 36-hour washing process, followed by the manual removal of the upper sheath. The sheath's bonded fibers were delicately removed before the foundational treatment. The foundational treatment entailed the removal of dirt and debris from the left work sheath. Upon completion of the initial treatment process, the fibers were dried in an oven and then subjected to chemical treatment, which was a 5% alkali treatment using NaOH for 3 hours (Manickaraj *et al.* 2022).

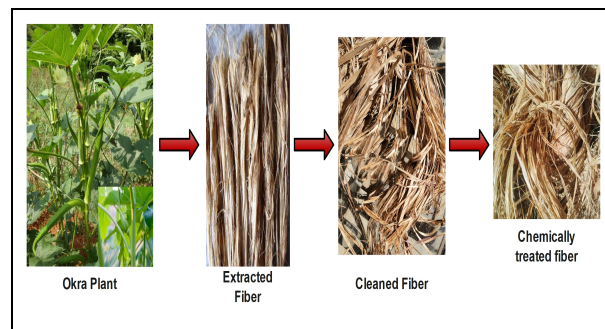


Fig. 1: Various stages of okra fiber extraction

The planar nature of these fibers meant that their breadth, not their diameter, could be utilized to measure thickness. The measurement was made consistent by implementing a unique gradation system. To sort the fibers by thickness, a rectangular slit was cut with slits measuring 1 to 3 mm at 1 mm interval. To ensure accurate categorization for use in the composite structure, the fibers were removed from the larger side. To ensure high quality, a Trinocular Vertical Microscope was used to measure the dimensions of five fiber samples, carefully chosen from each thickness category. Without having to measure each fiber individually, this approach has guaranteed precise fiber dimensions for the reinforcement procedure.

Depending on their thickness, the fibers were classified as 1 mm, 2 mm, and 3 mm after treatment. Then, the materials were cut into sections that were 6 mm to 18 mm long. Fig. 1 shows the steps involved in extracting okra fibers, which begin with collecting the plant, then move on to cleaning and chemical treatment, and culminate in sorting the fibers into specific lengths and cutting them into pieces. The hand layup technique was employed to fabricate the okra fibers/epoxy (OKFE) composite. Okra fibers that had been chemically treated were used as reinforcement. The weight percentages, lengths, thicknesses, and nSiO₂ concentrations of the fibers varied, with some being 4 wt.%, 8 wt.%, and 12 wt.%. The lengths ranged from 6 mm to 18 mm, and the thicknesses were 1 mm, 2 mm, and 3 mm. An epoxy matrix was mixed with SiO₂ nanoparticles (1, 2, and 3 wt.%) and then stirred mechanically for 15 to 30 minutes (Arumugam and Kaliannagounder, 2024). This augments the interaction between nSiO₂ and the polymer, resulting in enhanced characteristics. The fibers were subsequently included in the epoxy-nSiO₂ mixture to create the composite polymer matrix. Afterwards, a mold cavity was filled with the slurry and left to cure at room temperature for 24 hours so that it would solidify evenly.

There were 27 samples produced, each with a unique fiber weight, length, thickness, and nano-SiO₂ content. These samples were examined concerning their physical and mechanical parameters, including water absorption rate, flexural strength, and tensile strength.

A thorough assessment of the composite materials' performance is facilitated by the examination of mechanical parameters, such as flexural strength (FS) and tensile strength (TS). Mechanical properties, including tensile strength, were assessed through tensile testing, which involves the application of a progressively increasing axial load to the specimen until failure occurs. A Universal Testing Machine was used to measure factors like strain, displacement, and load. The production process for the OKFE composite specimen follows the specifications laid down by ASTM D 3039. Eq. 1 was employed to obtain the tensile strength.

$$\text{Tensile Strength (MPa)} = \frac{F_m}{A_c} \quad \dots (1)$$

where, A_c is the cross-sectional area, and F_m is the measured force.

The bending properties of the OKFE composites were evaluated through the flexural test. The specimen was made following the ASTM D790 criteria for the flexural test. The experiment employed a Universal Testing Machine (UNITEK 9940) to quantify

characteristics such as deflection and load. Eq. 2 was used to obtain the flexural strength.

$$\text{Flexural Strength (MPa)} = \frac{F_m L}{2wd^2} \quad \dots (2)$$

where, L represents length, w denotes width, F_m represents the measured force, and d denotes the thickness of the sample.

Following the guidelines laid out by ASTM D570-98, the moisture absorption test for the OKFE composite was done. Submerging the OKFE composite in water for 30 days allowed for periodic monitoring of the sample's weight. Eq. 3 was employed to ascertain the additional water absorption rate, which is expressed as a percentage.

$$W_A(\%) = \left(\frac{W_f - W_i}{W_i} \right) \times 100 \quad \dots (3)$$

where, W_f represents the final weightage, W_i represents the initial weightage of the specimen, and W_A , the water absorption percentage.

Table 1. Experimental input factors and their levels

Parameter	Parameter Code	Level		
		1	2	3
Weight percentage of OK fiber (wt.%)	W	4	8	12
Length of OK fiber (mm)	X	6	12	18
Thickness of OK fiber (mm)	Y	1	2	3
Nano-SiO ₂ content (wt.%)	Z	1	2	3

A methodical approach to experimental design was introduced by Taguchi, who prioritized the application of well-defined criteria. Being an approach based on line graphs, this method makes use of orthogonal arrays. The experimental settings and levels can be shown in Table 1. Also, whether creating models or optimizing existing ones, RSM is often used. RSM is intended to facilitate the efficient interaction of the chosen process factors. The experimental design matrix is followed through with the methodical execution of experiments, which are subsequently categorized using conventional classification methods.

2.3 Procedure for Implementing FFANN

Step 1: Preparation of Data

Results from 27 different permutations of factor W to factor Z were collected experimentally. In addition, the data was organized into two matrices: A, which

contains the input factor, and B, which comprises the range of the output properties.

Step 2: Normalizing Data

Matrices A and B were standardized to a uniform level to guarantee consistency in training.

Step 3: Splitting of Data

The data was split into two sets: one for training with 18 samples and another for testing with 9 samples.

Step 4: Initializing Network

The Feedforward Artificial Neural Network (FFANN) with one hidden layer, including 27 nodes, was initialized. The Adam optimizer was used for training, and the hidden layer was activated with the ReLU function. With a learning rate of 0.1, initialization can be started.

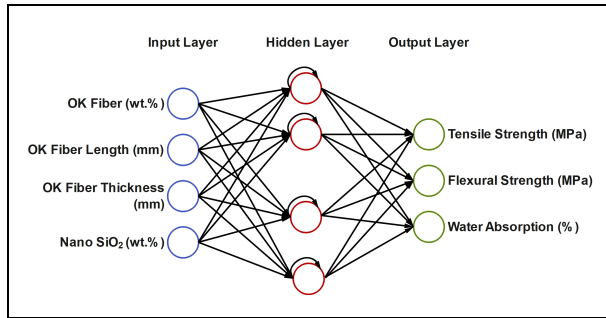


Fig. 2: A schematic illustration of the FFANN algorithm

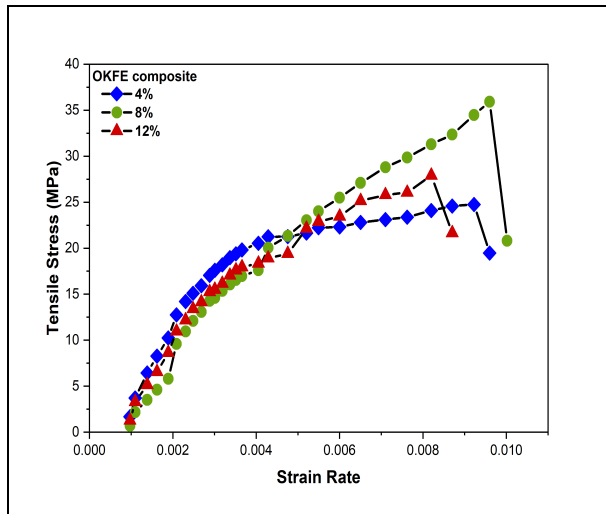


Fig. 3: Tensile stress vs. Strain of OKFE composite

Step 5: Training the Model

In order to determine the outputs, Forward Propagation involves sending the training data throughout the network. The loss computation involves finding the difference between the expected and actual outputs. The loss function can be reduced by modifying the network's weights and biases through the backpropagation process. Training was conducted on the model for 100 epochs through repetition.

Step 6: Testing the Model

This step involves predicting future property values by applying the training model to the test data. Finally, measurements like Mean Squared Error (MSE) were used to compare expected values with actual values in order to evaluate the model's performance.

Step 7: Interpretation

The outcome was examined to comprehend the impact of OK fiber weight percentage, length, thickness, and nSiO₂ on the anticipated qualities. Ultimately, the model may require change or fine-tuning to enhance predictive accuracy. The algorithm for an FFANN is illustrated in Fig. 2.

3. RESULTS AND DISCUSSION

Using a Taguchi L27 orthogonal array, this section assesses the mechanical and physical properties of the OKFE composite, such as TS, FS, and water absorption (WA).

3.1 Tensile Properties

Fig. 3 illustrates the tensile strain variation in fibers with varying weight percentages. The maximum tensile strength was observed in the OK fiber at a concentration of 8 wt.%. The 12 wt.% OK fibers were susceptible to fracture and manifested a low tensile strength. Fig. 4 (a) and (b) illustrate the TS with varying fiber weight percentage with respect to fiber length and fiber thickness, respectively. These graphs revealed that the maximal value of tensile strength was achieved by the OKFE composite with a 12 mm OK fiber length. The optimal length of the OK fiber is 12 mm.

Furthermore, as shown in Figures 4 (c) and (d), the OK fiber thickness has a clear impact on the TS of the OKFE composite. The 2 mm is the optimum for OK fiber thickness. These findings show that the OK fibers' thickness and length slightly affect the tensile strength of the OKFE composites. Additionally, the study showed that out of the three tested composites, the 8 wt.% OKFE composites had the highest tensile strength, followed by the 12 wt.% OKFE composites. The ultimate tensile strength of a composite of 8 wt.% OK fibers with 12 mm long and 2 mm thick dimensions is 35.92 MPa. Similarly, composites made from Indian almond fibers showed the best tensile strength (58 MPa) at 10 mm fiber length. Reinforcing with 10 wt.% Indian jujube seed particles further improved tensile, flexural, and impact strength by 14–23% (Senthilkumar *et al.* 2023). Owing to the significant improvement in tensile strength, the bonding between the epoxy matrix and OK fiber is greatly enhanced. The 8 wt.% OK fibers in the 2 mm-thick and 12 mm-long OKFE composite have satisfactory bonding and tensile strength. While 8 wt.% OKFE fibers with 6 mm and 18 mm lengths are slightly stronger than fibers with 12 mm lengths, they are still not quite as strong. OK fibers with a 2 mm thickness have an improved tensile strength that is slightly lower than that of 8 wt.% fibers with a 1 mm or 3 mm thickness.

The ideal parameters for OKFE composites are 2 mm thickness, 12 mm length, and 8 wt.% fibers, which yields a maximum tensile strength of 35.92 MPa. Because lower weight percentages (like 4 wt.%) are not reinforcing enough and because 12 wt.% can cause fiber aggregation and decreased interfacial bonding, 8 wt.% strikes an optimum. The 12 mm fiber length optimizes load-bearing capacity in contrast to shorter (6 mm) and longer (18 mm) fibers, which may experience misalignment and insufficient engagement with the matrix. The surface area for bonding is optimized by the

specified thickness of 2 mm. Thin fibers (1 mm) are unable to withstand tensile pressures, while thicker fibers (3 mm) introduce tension concentration points.

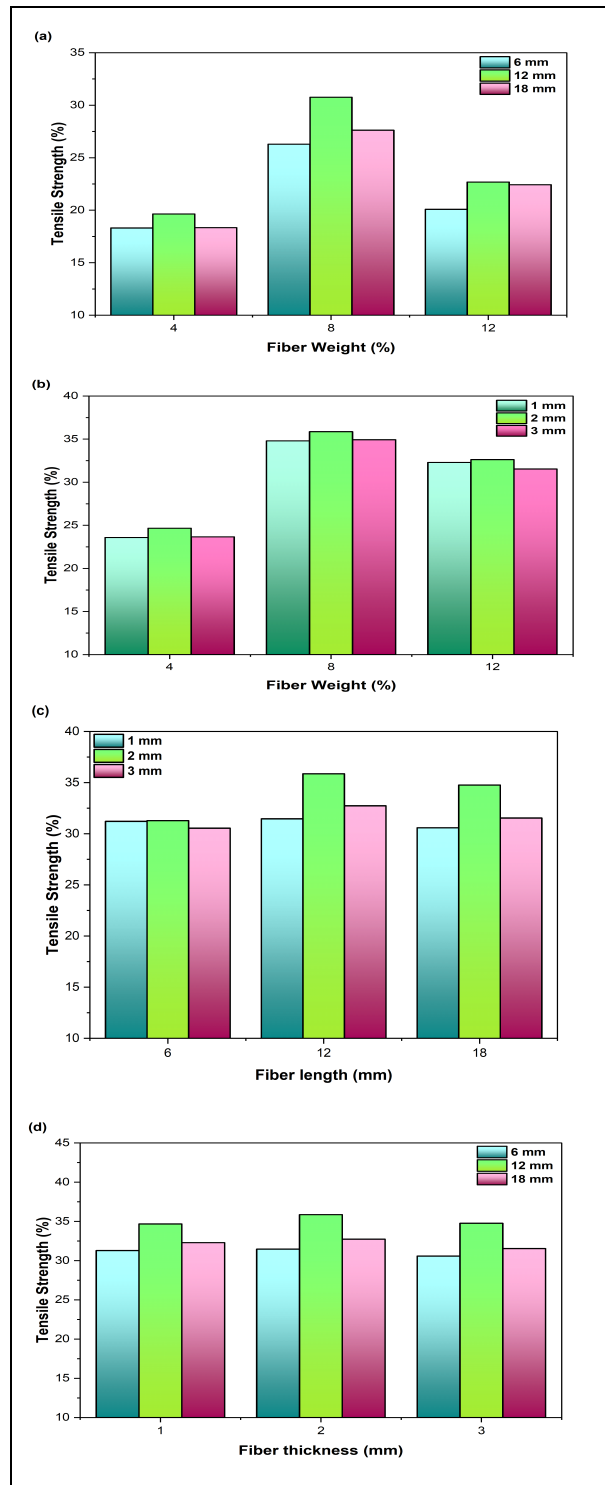


Fig. 4: Evaluation of TS with (a) different fiber weight and fiber length at a constant fiber thickness of 2mm, (b) different fiber weight and fiber thickness at a constant fiber length of 12mm, (c) different fiber length and fiber thickness at constant fiber weight of 8wt% and (d) different fiber thickness and fiber length at constant fiber weight of 8wt%.

A variety of underlying mechanisms that are associated with the composite structure and fiber-matrix interactions can be attributed to the tensile strength variations in the OKFE composite, which are influenced by the W to Z factor. The optimal weight percentage of fibers is 8 wt.%, as it achieves an equilibrium between maintaining adequate matrix continuity and providing sufficient fiber reinforcement. The stress transfer efficiency is enhanced by the fibers' ability to effectively bear the load and maintain excellent dispersion at this concentration. At maximum fiber weight percentages (12 wt.%), the composite can undergo stress concentration spots and early failure due to fiber aggregation and insufficient interfacial bonding. The optimal fiber length is 12 mm, because it maximizes the load-bearing capacity, decreases the risk of fiber pull-out or breakage (6 mm), and avoids problems with misalignment and inefficient stress distribution (18 mm). A 2 mm fiber thickness ensures that the surface area is optimal for bonding without compromising flexibility or causing undue rigidity. Thinner fibers (1 mm) have insufficient reinforcement to effectively sustain the tensile pressures. Conversely, the potential for matrix fracture and stress concentration is the consequence of denser fibers (3 mm).

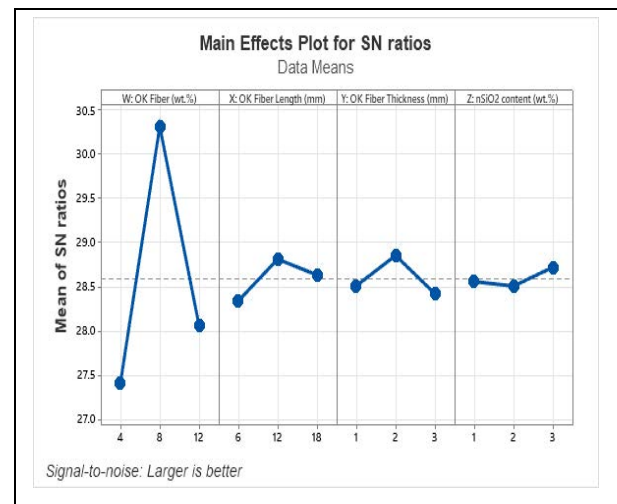


Fig. 5: Effect of process parameters for tensile strength (S/N ratios)

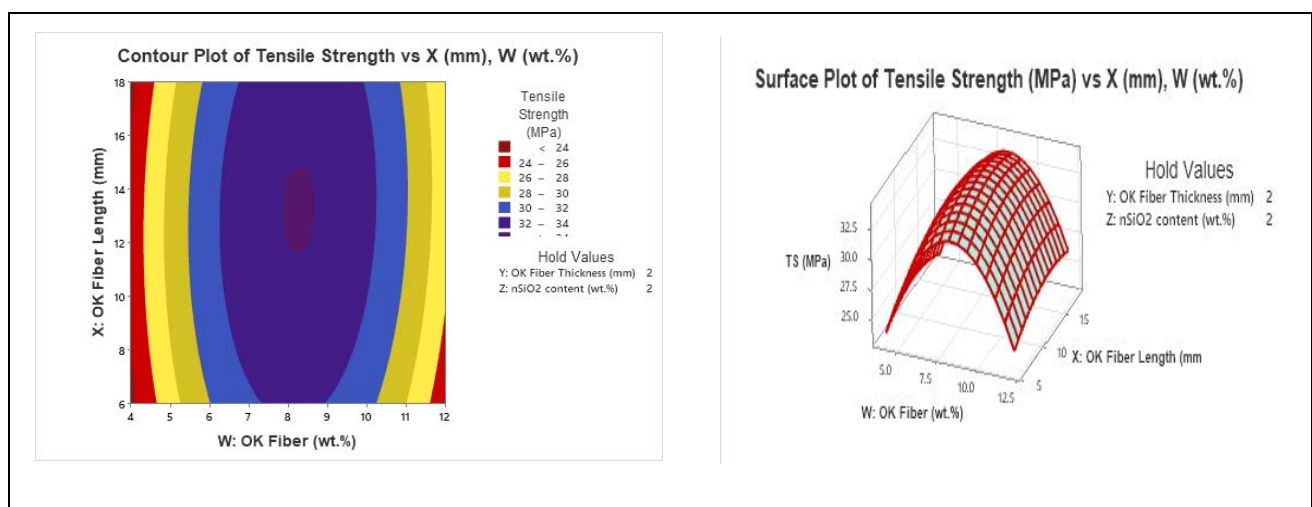
Fig. 5 depicts the process parameter plot of S/N ratios for TS, which reveals that the best grouping for maximal TS is 8 wt.% OK fibers with 12 mm fiber length, 2 mm thickness, and 3 wt.% nSiO₂ content. Table 2 shows that the OK wt.% has the greatest impact on tensile strength (89.85%), OK fiber length (2.61%), OK fiber thickness (1.78%), and nSiO₂ content (0.45%). Furthermore, the fiber length and fiber weight percentage combination is substantial at 4.39%. The statistical significance of the remaining combinations is negligible. Using the computed regression coefficients, which are shown in Table 3, a second-degree equation (Eq. 7) is created to forecast the TS for the selected parameter range (Ponnuswamy *et al.* 2024).

Table 2. ANOVA Results for S/N Ratios

Source	DF	Seq SS	Contribution	Adj MS	F Value	P Value
W (wt.%)	2	443.357	89.85%	221.678	1025.21	0.000
X (mm)	2	12.880	2.61%	6.440	29.78	0.001
Y (mm)	2	8.783	1.78%	4.392	20.31	0.002
Z (wt.%)	2	2.217	0.45%	1.109	5.13	0.050
W (wt.%) * X (mm)	4	19.138	3.88%	4.785	22.13	0.001
W (wt.%) * Y (mm)	4	3.772	0.76%	0.943	4.36	0.054
W (wt.%) * Z (wt.%)	4	1.990	0.40%	0.498	2.30	0.173
Error	6	1.297	0.26%	0.216		
Total	26	493.435	100.00%			

Table 3. Statistical estimation of regression coefficients for TS

Term	Coeff.	SE Coeff.	T Value	P Value
Constant	-6.53	5.26	-1.24	0.237
W (wt.%)	8.400	0.631	13.31	0.000
X (mm)	0.640	0.433	1.48	0.163
Y (mm)	3.95	3.32	1.19	0.256
Z (wt.%)	-2.51	2.54	-0.99	0.342
W (wt.%) * W (wt.%)	-0.5277	0.0334	-15.80	0.000
X (mm) * X (mm)	-0.0339	0.0148	-2.29	0.040
Y (mm) * Y (mm)	-1.157	0.617	-1.88	0.083
Z (wt.%) * Z (wt.%)	0.632	0.617	1.02	0.324
W (wt.%) * X (mm)	0.0176	0.0157	1.12	0.283
W (wt.%) * Y (mm)	-0.0325	0.0944	-0.34	0.737
W (wt.%) * Z (wt.%)	0.0646	0.0944	0.68	0.506
X (mm) * Y (mm)	0.064	0.103	0.62	0.545
X (mm) * Z (wt.%)	-0.008	0.103	-0.08	0.935

**Fig. 6: 2D plot and 3D plot for TS**

$$\text{Tensile Strength (MPa)} = -6.53 + 8.400 W + 0.640 X + 3.95 Y - 2.51 Z - 0.5277 W^2 - 0.0339 X^2 - 1.157 Y^2 + 0.632 Z^2 + 0.0176 W * X - 0.0325 W * Y + 0.0646 W * Z + 0.064 X * Y - 0.008 X * Z \quad \dots (7)$$

The OK fiber thickness is maintained at 2 mm, resulting in the production of the contour (2D) and surface (3D) graphs (Fig. 6). The figure indicates that the optimal TS can be reached by adjusting the OK length to 6 mm–12 mm and the fiber weight percentage to 7.5%–8.5%. In addition, it is evident that the TS is saturated and then falls with increasing percentage and fiber length.

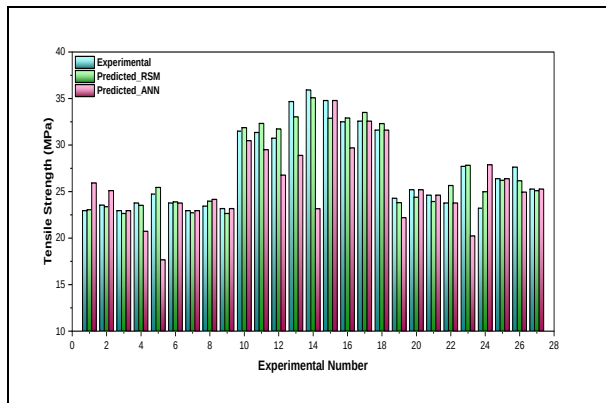


Fig. 7: Experimental TS evaluation with RSM and ANN estimations

Experimental findings for TS, along with the predictions from RSM and ANN, are shown in Fig. 7. The average error of ANN predictions is 28.41%, while that of RSM predictions is 24.22%.

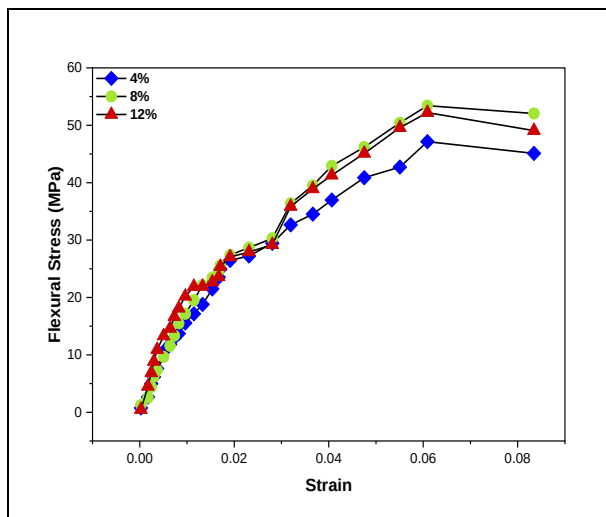


Fig. 8: Evaluation of Flexural Strength of stress vs. strain of composites for different fiber wt.%.

3.2 Flexural Properties

Fig. 8 represents the stress vs. strain graph of the OKFE composites with different weight percentages of

epoxy-reinforced fiber when it is bent or flexed. At 8 wt.% OK fiber, the OKFE composite shows the best flexural strength. The OKFE composite's flexural strength was reduced due to fiber agglomeration, which was brought about by both high and low fiber concentration. The flexural testing of the OKFE composite revealed its stress and deformation behavior.

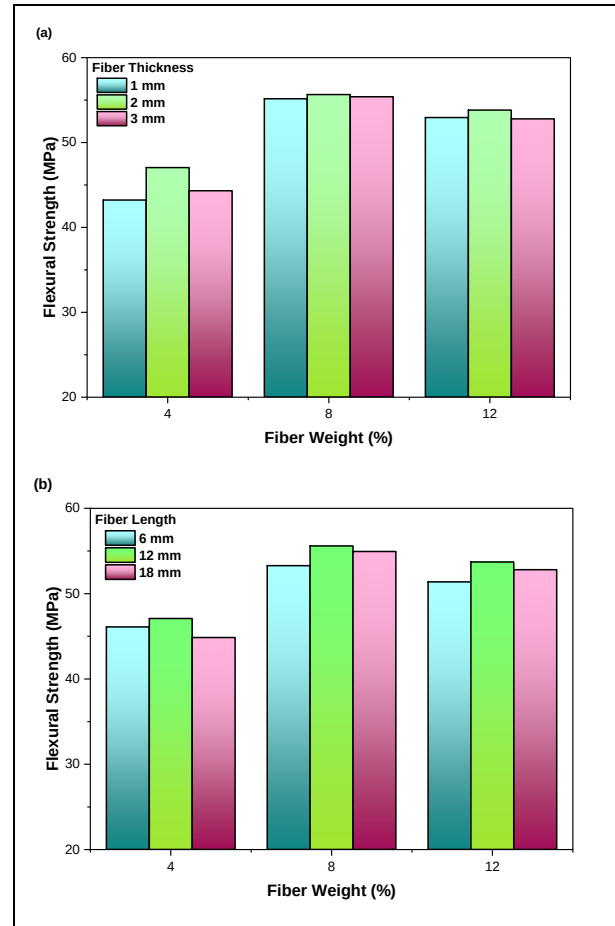


Fig. 9: Evaluation of flexural strengths when: (a) fiber thickness = 2 mm, and (b) fiber length = 12 mm

As illustrated in Fig. 9, the thickness of the OK fiber has only a minor impact on the composite's flexural strength. 2 mm is the optimal thickness for OKFE composites in terms of FS. As the fiber content rises and the fiber thickness decreases, the void content of the material rises dramatically. The FS of the OKFE composites is reduced due to the existence of cavities. The FS decreases as the fiber content increases because the epoxy and fibers become less adherent to one another. The length of the fibers is slightly increased, which marginally enhances the FS of the composites. The length of the fibers enhances their reinforcement. From Fig. 7, it is evident that the optimal length of the OK fiber is 18 mm. The void content was also exacerbated by the excessive length of the fibers, which complicated the stir casting process. As a result, the flexural stiffness of the material is reduced. The highest level of flexural strength

was shown by the epoxy composite that was 2 mm thick and 12 mm long and reinforced with OK fibers. The flexural strength of the composites is usually proportional to the fiber weight percentage. An OKFE composite with 8 wt.% OK fiber is ideal. An ultimate TS of 55.65 MPa is attained by the OKFE composites.

Because they significantly affect the stress distribution and structural integrity of the material, the FS of the OKFE composite is mainly governed by the fiber content, width, length, and nano SiO₂ concentration. The composite attains its maximum flexural strength at a fiber content of 8 wt.%. This concentration ensures that the matrix cohesion and fiber reinforcement remain in a stable equilibrium, allowing effective stress transmission without the composite-compromising agglomeration of large amounts of fibers. In contrast, FS decreases with increasing and decreasing fiber weight percentages, respectively. Fiber clustering and an increase in void content are more likely to occur with higher percentages, while insufficient reinforcing is the only benefit of lower percentages. The best specimen between reinforcing and void minimization is the one with 2 mm of fiber thickness. Fibers having a width of less than 2 mm enhance the void content, whereas fibers with greater than 2 mm diminish the bond between the fiber and epoxy. Both of these variables have a detrimental effect on the fiber's flexural strength. The optimal flexural strength of 12 mm fibers is initially demonstrated; however, the 18 mm length may introduce additional voids and confound the manufacturing process, resulting in a reduction in overall strength. This is due to the

improved load distribution along the fiber. Therefore, the flexural strength of the OKFE composites is guaranteed to be at the highest level by the optimum of 8 wt.% fiber content, 2 mm thick, 12 mm length, and 1 wt.% n-SiO₂. This is achieved by minimizing voids, optimizing stress distribution, and preserving a robust fiber-matrix interface. Similarly, Nadendla and Kopparthi (2022) synthesized nanocellulose from okra fiber through chemical and minor mechanical methods to fabricate an advanced composite laminate with cotton fabric (CF) reinforcement. The hybrid composites attained a flexural strength of 116.48 MPa and a modulus of 4.21 GPa, while pure nanocellulose-reinforced composites reached the highest strength.

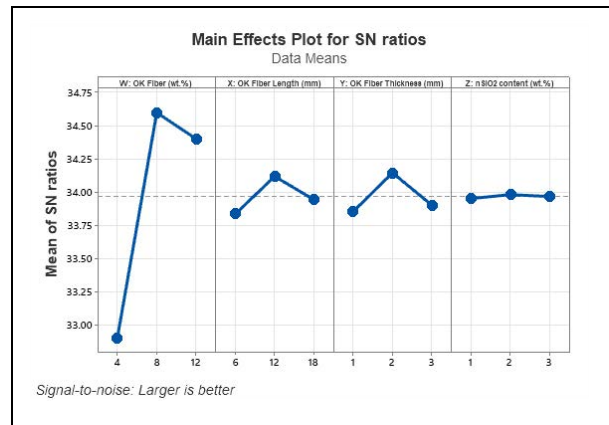


Fig. 10: Effect of process parameters for FS (S/N ratio)

Table 4. Results on ANOVA for FS (S/N ratios)

Source	DF	Seq SS	Contribution	Adj MS	F Value	P Value
W (wt.%)	2	485.355	92.04%	242.678	1841.04	0.000
X (mm)	2	12.911	2.45%	6.456	48.97	0.000
Y (mm)	2	13.200	2.50%	6.600	50.07	0.000
Z (wt.%)	2	0.131	0.02%	0.065	0.50	0.632
W (wt.%) * X (mm)	4	8.887	1.69%	2.222	16.86	0.002
W (wt.%) * Y (mm)	4	5.802	1.10%	1.451	11.00	0.006
W (wt.%) * Z (wt.%)	4	0.263	0.05%	0.066	0.50	0.740
Error	6	0.791	0.15%	0.132		
Total	26	527.340	100.00%			

Table 5. Statistical estimation of Regression coefficients for FS

Term	Coeff.	SE Coeff.	T Value	P Value
Constant	13.97	4.05	3.45	0.004
W (wt.%)	6.312	0.485	13.01	0.000
X (mm)	0.747	0.333	2.24	0.043
Y (mm)	7.28	2.55	2.85	0.014
Z (wt.%)	0.05	1.96	0.03	0.978

Term	Coeff.	SE Coeff.	T Value	P Value
W (wt.%) * W (wt.%)	-0.3369	0.0257	-13.12	0.000
X (mm) * X (mm)	-0.0371	0.0114	-3.25	0.006
Y (mm) * Y (mm)	-1.627	0.474	-3.43	0.004
Z (wt.%) * Z (wt.%)	-0.172	0.474	-0.36	0.723
W (wt.%) * X (mm)	0.0163	0.0121	1.34	0.202
W (wt.%) * Y (mm)	-0.0390	0.0726	-0.54	0.600
W (wt.%) * Z (wt.%)	0.0007	0.0726	0.01	0.993
X (mm) * Y (mm)	-0.0165	0.0791	-0.21	0.838
X (mm) * Z (wt.%)	0.0521	0.0791	0.66	0.522

Fig. 10 shows the process parameter plot of S/N ratios for FS. The optimum grouping for maximal flexural strength was found to be 8 wt.% OK fibers, with 12 mm fiber length, 2 mm thickness, and 2 wt.% nSiO₂. The Analysis of Variance (Table 4) shows that the OK wt.% has the highest impact on flexural strength (92.04%). Next on the list of important criteria are the OK fiber thickness (2.50%), length (2.45%), and n-SiO₂ (0.02%). In addition, there is a combination of 1.69% for fiber length and weight 1.10% for fiber length and thickness. The other permutation has a statistically insignificant impact. The computed regression coefficients are shown in Table 5, and they are used to construct the second-degree Eq. 8 that predicts the FS for the given parameter range.

$$\text{Flexural Strength (MPa)} = 13.97 + 6.312 W + 0.747 X + 7.28 Y + 0.05 Z - 0.3369 W^2 - 0.0371 X^2 - 1.627 Y^2 - 0.172 Z^2 + 0.0163 W * X - 0.0390 W * Y + 0.0007 W * Z - 0.0165 X * Y + 0.0521 X * Z \quad \dots (8)$$

By preserving the thickness of the OK fiber at 2 mm, the 2D and 3D diagrams (Figure 11) are generated. The figure indicates that the optimal FS can be reached by adjusting the length of fiber to 9.5 mm—13 mm and the fiber weight percentage to 8%—11%. Furthermore, it is clear that the FS reaches saturation and subsequently decreases as the fiber length and percentage increase.

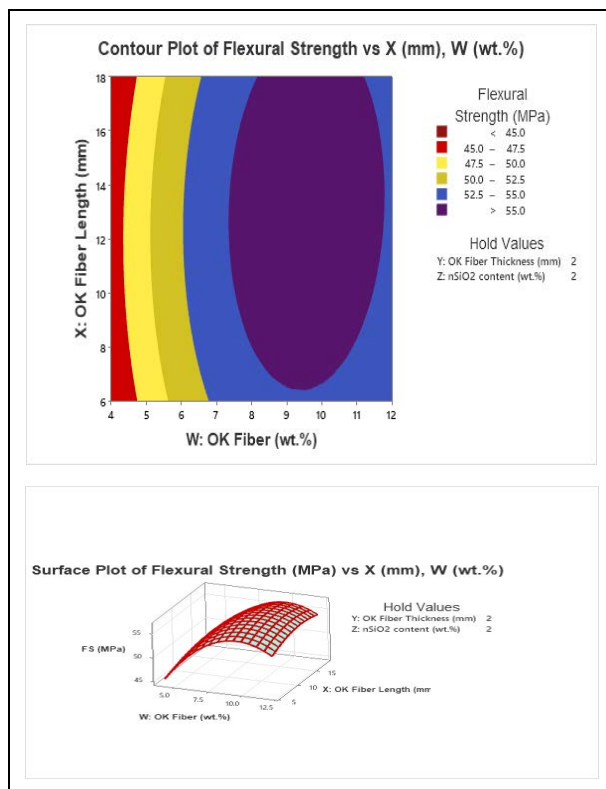


Fig. 11: 2D plot and 3D plot for FS

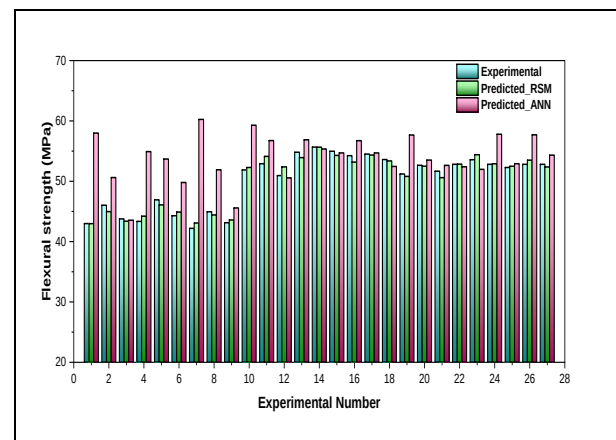


Fig. 12: Experimental FS evaluation with RSM and ANN predictions

The experimental results of FS were compared graphically with predictions using RSM and ANN in Fig. 12. A comparison of ANN and RSM prediction accuracy yields a difference of 6.71% vs. 0.015%, respectively. Likewise, the hybrid composites made of jute, kenaf, and glass fibers with aluminum oxide (Al₂O₃) nanoparticles were studied. They applied RSM and ANN to optimize fiber orientation, sequence, and nanoparticle content, and found that ANN gave better prediction accuracy. Optimal mechanical properties were achieved at 90° fiber orientation, 3 fiber layers, and 5% Al₂O₃ content (Arunachalam *et al.* 2025).

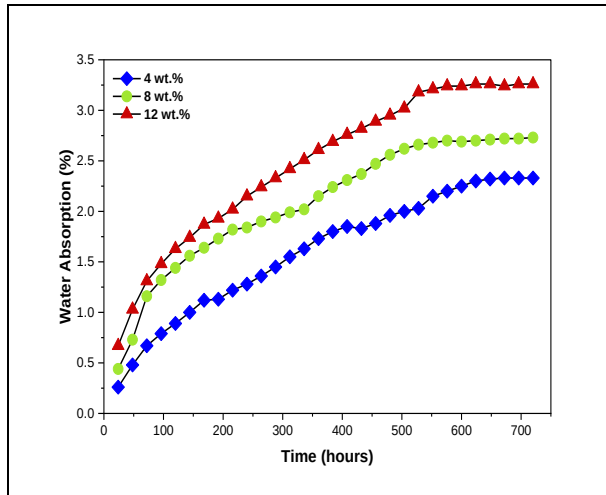


Fig. 13: Evaluation of WA for different weight percentages of fiber

3.3 Water Absorption

The inherent hydrophilicity of the fibers causes a small rise in the WA of the OKFE composite with increasing fiber content. The rate of water absorption over time is depicted in Fig. 13. Within 30 days, the OKFE composite with a 20 wt.% demonstrated a maximal water absorption rate of 3.26%. Extensive research shows that different fiber weight percentages (wt.%) affect the OKFE composite's water absorption rate. With 8 wt.% OKFE, the composite has a nominal water absorption rate of 2.73%. At first, the rate of water absorption is very high. It has steadied at a constant level after 450 hours. The WA rate of OKFE composites increases with increasing fiber weight percentage because OK fibers are hydrophilic, implying they absorb and hold moisture. The 30 days of data collection from the 12 wt.% OKFE composites showed an absorption rate of 3.26%, demonstrating this aspect. Fibers with a high weight percentage are better at absorbing water because they have a higher surface area to expose and more paths

for water to enter. On the other hand, the absorption rate of the 8 wt.% OKFE composite is 2.73%, which is slightly lower than the other two. This suggests that the mechanical properties are improved without there being an overwhelming increase in water intake. This is supported by the sufficient fiber content. The material's rapid initial water absorption rate, which stabilizes after 450 hours, indicates that it absorbs water at a high rate until it reaches a saturation limit. This was observed in all composites. After protracted exposure, the steady state is indicative of a consistent water absorption level, which is the result of saturation. The presence of water droplets on the surface of the composite material suggests that water has penetrated and become trapped at the contact between the fibers and the matrix. Since the fibers get more hydrophilic and interact with water more effectively as the fiber concentration increases, the water absorption rate rises as well. In contrast, when absorption levels remain relatively constant over time, it means that all of the available hydrophilic sites in the composite have been used up.

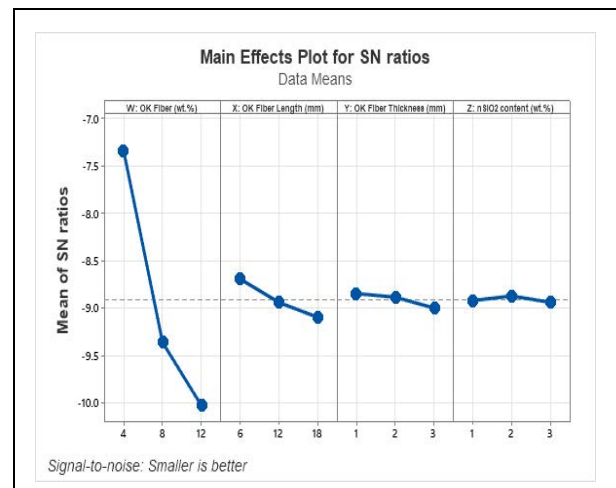


Fig. 14: The process parameter plot of WA (S/N ratios)

Table 6. ANOVA for SN ratios

Source	DF	Seq SS	Contribution	Adj MS	F Value	P Value
W (wt.%)	2	3.44048	96.43%	1.72024	2060.03	0.000
X (mm)	2	0.07861	2.20%	0.03930	47.07	0.000
Y (mm)	2	0.01247	0.35%	0.00624	7.47	0.024
Z (wt.%)	2	0.00287	0.08%	0.00143	1.72	0.257
W (wt.%) * X (mm)	4	0.01986	0.56%	0.00497	5.95	0.028
W (wt.%) * Y (mm)	4	0.00228	0.06%	0.00057	0.68	0.630
W (wt.%) * Z (wt.%)	4	0.00639	0.18%	0.00160	1.91	0.228
Error	6	0.00501	0.14%	0.00084		
Total	26	3.56796	100.00%			

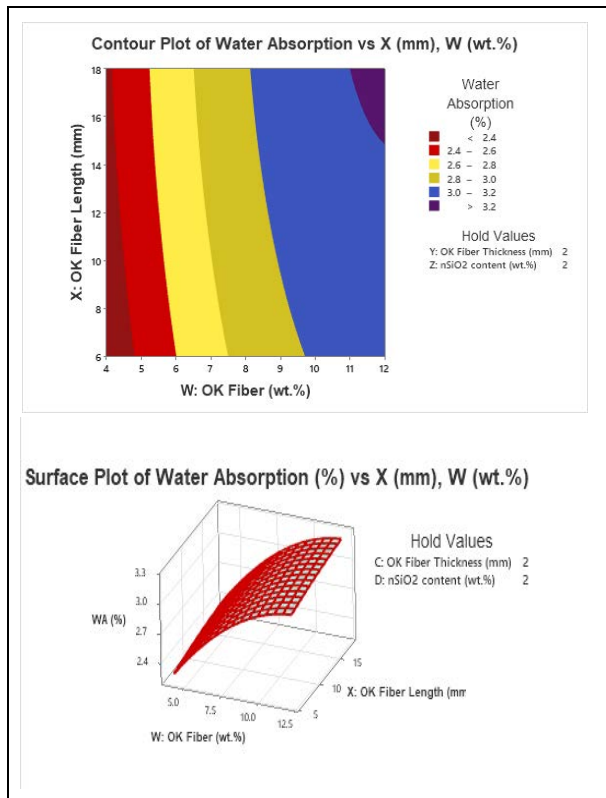
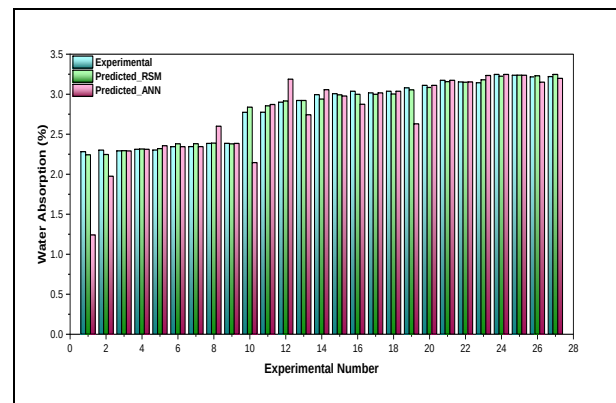
Table 7. Statistical estimation of Regression coefficients for Water Absorption

Term	Coeff.	SE Coeff.	T value	P value
Constant	1.179	0.192	6.13	0.000
W (wt.%)	0.2865	0.0231	12.42	0.000
X (mm)	0.0264	0.0158	1.67	0.120
Y (mm)	0.012	0.121	0.10	0.925
Z (wt.%)	-0.0669	0.0930	-0.72	0.485
W (wt.%) * W (wt.%)	-0.01184	0.00122	-9.70	0.000
X (mm) * X (mm)	-0.000451	0.000542	-0.83	0.420
Y (mm) * Y (mm)	0.0118	0.0225	0.52	0.609
Z (wt.%) * Z (wt.%)	0.0097	0.0225	0.43	0.675
W (wt.%) * X (mm)	0.000190	0.000575	0.33	0.747
W (wt.%) * Y (mm)	0.00153	0.00345	0.44	0.665
W (wt.%) * Z (wt.%)	0.00174	0.00345	0.50	0.622
X (mm) * Y (mm)	-0.00368	0.00376	-0.98	0.345
X (mm) * Z (wt.%)	0.00060	0.00376	0.16	0.875

The optimal combination for achieving the lowest water absorption rate is 12 wt.% OK fibers with 18 mm fiber length, 3 mm thickness, and 3 wt.% n-SiO₂, as shown in Fig. 14, which illustrates the processing factor plot of S/N ratios for WA rate. Table 6 clearly shows that the OK wt.% is the major factor controlling the WA rate at 96.43%. The n-SiO₂ concentration, OK fiber thickness, and OK fiber length are 0.08%, 0.35%, and 2.20%, respectively. Additionally, at 0.56%, the composite of fiber weight percentage and fiber length plays a significant role. The last pairings are linked with minimal statistical significance. Equation (11) is a second-degree equation that predicts the water absorption rate within the given parameter range. It is derived from Table 7, which provides the predicted regression coefficients.

$$\text{Water Absorption (\%)} = 1.179 + 0.2865 W + 0.0264 X + 0.012 Y - 0.0669 Z - 0.01184 W^2 - 0.000451 X^2 + 0.0118 Y^2 + 0.0097 Z^2 + 0.000190 W * X + 0.00153 W * Y + 0.00174 W * Z - 0.00368 X * Y + 0.00060 X * Z \dots (11)$$

By preserving the thickness of the OK fiber at 2 mm, the 2D and 3D plots (Fig. 15) are generated. By adjusting the OK fiber length to 6–8 mm and the fiber weight percentage to 8%, it is evident from the figure that the lowest water absorption rate can be attained. Furthermore, the rate of water absorption increases as the fiber percentage increases. Fig. 16 shows a visual comparison between the measured water absorption rate and the predictions made using RSM and ANN. RSM forecasts are 0.018% accurate on average, and ANN predictions are 2.74% accurate. The water absorption and diffusion behavior in HDPE biocomposites reinforced with *Washingtonia filifera* (WF) fibers were studied. Using RSM, ANN, and genetic algorithm optimization, they found that 10% fiber content and 1-hour immersion minimized water absorption following Fickian diffusion. ANN models provided excellent prediction accuracy, with correlation coefficients above 0.99 (Belaadi *et al.* 2024).

**Fig. 15: 2D and 3D plots for WA****Fig. 16: Experimental WA evaluation with RSM and ANN predictions**

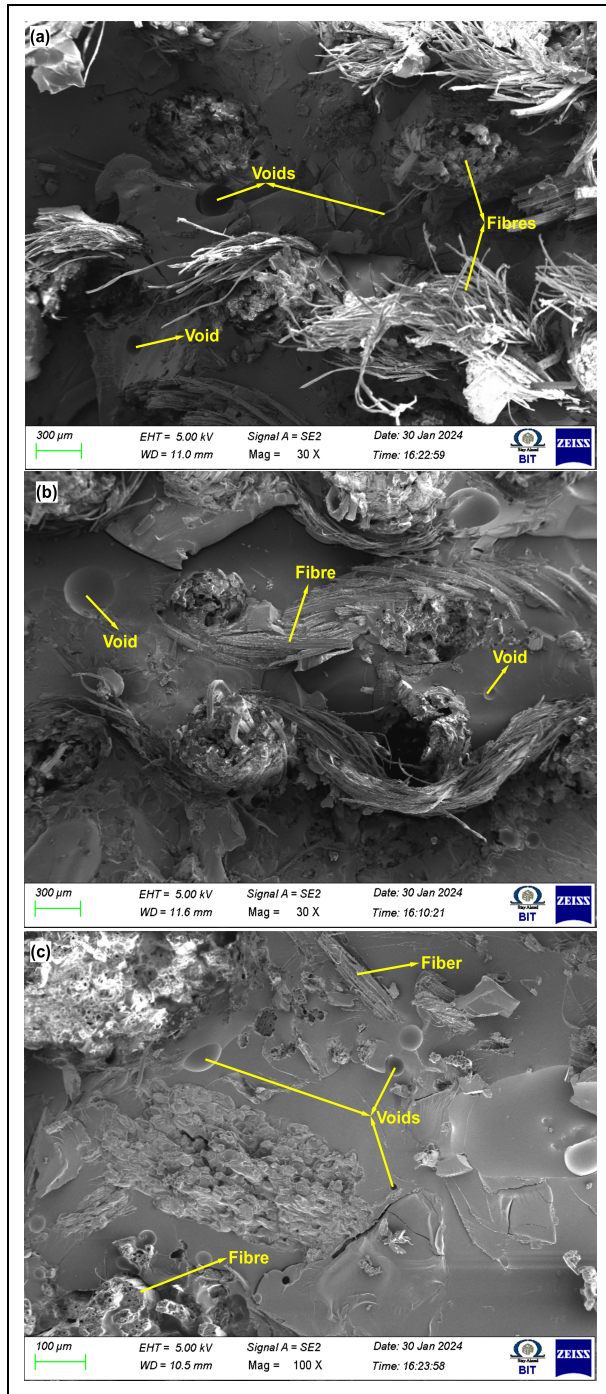


Fig. 17: Microstructure of OKFE composite (a) 4 wt.% OKFE fiber composites, (b) 8 wt.% fiber composites, and (c) 12 wt.% fiber composites

3.4 Microstructural Analysis

Microscopical examination of the OKFE composite reveals the effects of different fiber contents on the microstructural properties. As shown in Fig. 17 (a), the OKFE composite containing 4 wt.% OK fibers was first examined. In this specimen, the void contents of the composite are clearly visible. These areas show that the epoxy has not been completely integrated with the fibers

or that air pouches were stuck during manufacture. These holes distort the composite's surface texture and introduce mechanical performance weaknesses by compromising the structural integrity of the material.

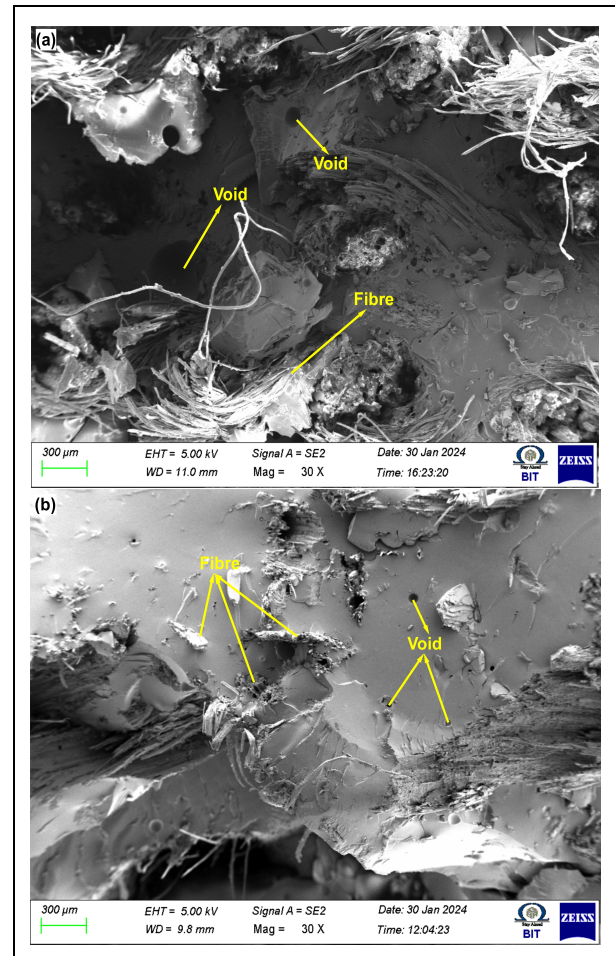


Fig. 18: SEM analysis of OKFE composite of (a) tensile sample and (b) flexural sample at 4 wt.% fiber content, 12 mm fiber length, 2 mm fiber thickness, and 2 wt.% n-SiO₂ fabricated composite

The microscopical analysis of the 8 wt.% OKFE composite is depicted in Fig. 17 (b). In comparison with the 4 wt.% specimen, the composite exhibits a substantial decrease in void content in this configuration. The epoxy matrix exhibits a more uniform distribution of fibers, which leads to a composite surface that is smoother. Reductions in surface roughness and cavities are signs of better impregnation and matrix-fiber bonding. This change likely enhanced the mechanical properties as a whole, including tensile and flexural strength. The sample's suitable fiber content results in a smooth surface, which avoids the production of air pockets and flaws while improving matrix penetration and fiber dispersion. Fig. 17 (c) shows that there is an increase in surface contaminants and void contents in the 12 wt.% OKFE composites. The fiber clusters have an uneven distribution within the matrix as a result of the increased

fiber concentration in certain areas, preventing the matrix from penetrating completely. The mechanical performance of the composites may be compromised due to the presence of these flaws and surface contaminants, which could result in a lack of homogeneity. At this greater percentage, fibers tend to cluster, which can make it harder for the matrix to circulate them. As a result, surface voids and flaws become more noticeable.

The SEM images of the OKFE composite of 4 wt.% fiber content, 12 mm fiber length, 2 mm fiber thickness, and 2 wt.% n-SiO₂ fabricated composite are depicted in Fig. 18, which offers important insights into the microstructural failure mechanisms and interfacial bonding. Fig. 18 (a) shows the tensile sample with fiber pull-out and void formation, and Fig. 18 (b) shows the flexural sample of better fiber-matrix bonding and reduced micro voids. SEM analysis demonstrates a strong bonding between the epoxy matrix and the okra fibers. The mechanical integrity and performance of the composites are contingent upon the efficient transmission of stress from the matrix to the fibers, as demonstrated by the well-bonded interface.

4. CONCLUSION

The performance of polymer composites reinforced with okra fibers evaluated in this study has provided valuable insights into their mechanical, physical, and microstructural characteristics. By improving the fiber content at 8 wt.%, 12 mm fiber length, fiber of 2 mm thickness, and 3 wt.% n-SiO₂ content, the study noted a notable improvement in multiple essential parameters. By finding the optimum between matrix continuity and fiber reinforcement, the ideal arrangement allowed for effective load transfer and a maximum tensile strength of 35.92 MPa. Fiber aggregation and poor interfacial bonding reduced the tensile strength of composites with a 12 wt.% fiber content, whereas inadequate reinforcement was given by those with a 4 wt.% fiber content. With 8 wt.% fibers, a 2 mm thickness, and 2 wt.% n-SiO₂ content, which prevented cavities and ensured the best possible adhesion between the matrix and fibers, the flexural strength reached 55.65 MPa. Additionally, the stress dispersion and load transmission were improved by the fiber length of 12 mm. At a fiber content of 12 wt.%, the optimal absorption rate was 3.26%, demonstrating that the hydrophilic characteristics of OK fibers boosted water absorption. The optimal 8 wt.% setup achieved a compromise between mechanical performance and moisture resistance with a lowered absorption rate of 2.73%. The accuracy of prediction models using Feed-Forward Artificial Neural Networks was also much higher than that of models using Response Surface Methodology. Tensile strength (24.22%), flexural strength (0.015%), and water absorption (0.018%) were the three average forecast errors. In the field of composite material research, FFANN's remarkable performance

highlights its ability to do complicated predictive analyses.

It is recommended that future studies explore the effects of humidity, ultraviolet light, and temperature cycling, among other environmental factors, on the composites' long-term durability. Composites that incorporate nanomaterials, such as graphene or carbon nanotubes, may have their mechanical, thermal, and barrier properties improved. It would be beneficial to enhance fiber-matrix bonding and decrease void content in manufacturing techniques, particularly at elevated fiber loadings, to make composites reinforced with okra fibers and n-SiO₂ perform better overall, expanding their use in different high-performance industries.

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CONFLICTS OF INTEREST

The authors declare that there is no conflict of interest.

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